

Introduction

The human mind is well adapted to idea generation and composition, being able to tackle complex problems. However, we are still vulnerable to biases, such as Time Estimation Bias and Time Blindness. These biases can impact our perception and management of time, affecting our productivity. According to Art Markman (2017), the concept of time plays a crucial role in how we perceive and remember events. He introduces the Construal Level Theory, when we view a landscape, we perceive it as a whole, but when we hold an apple, we notice its intricate details, such as its moistness. This theory suggests that as the distance (time) between us and an event increases, our idea or concept of that event becomes vaguer and less structured.

This phenomenon has significant implications for long-term projects, where maintaining focus and structure over an extended period can be challenging. Thus, the student's aim to address this challenge by developing a structured tool for managing projects, specifically tailored for architectural projects, which often span weeks, months, or even years.

The proposed system introduces a hierarchical structure to project management, consisting of Projects, Deliverables, and Tasks. This structured approach aims to provide a clear framework to help individuals and teams stay organized, focused, and efficient throughout the entire project lifecycle.

user_info		project_info		deliverable		task	
user_id	integer	project_id	integer	deliverable_id	integer	task_id	integer
username	varchar	title	varchar	project_id	integer	deliverable_id	integer
email	varchar	details	text	title	varchar	project_id	integer
password	varchar	image_dir	varchar	details	text	title	varchar
user_type	int	status	varchar	status	varchar	details	text
first_name	varchar	created_at	timestamp	created_at	timestamp	status	varchar
last_name	varchar	due_date	date	due_date	date	created_at	timestamp
cp_number	varchar	client_name	varchar	completion_date	date	due_date	date
date_of_birth	date	client_email	varchar	review_status	varchar	completion_date	date
address	varchar	qr_code	varchar	assigned_to	integer	review_status	varchar



Figure 1.1 Entity Relationship Diagram

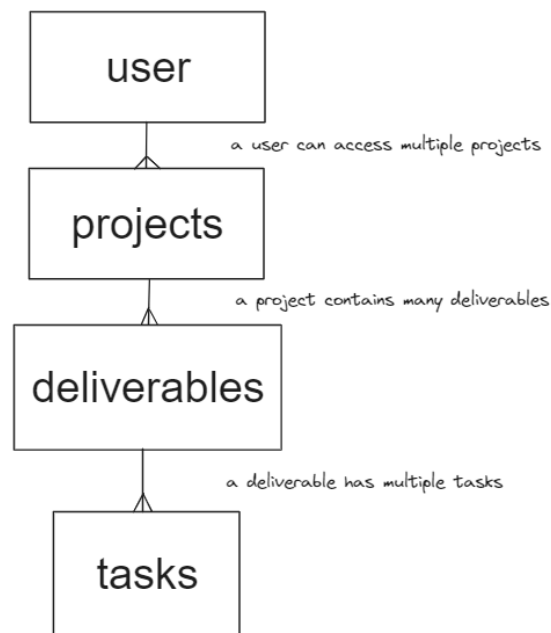


Figure 1.2 Concept Diagram

Table	Action	Rows	Type	Collation	Size	Overhead
☐ deliverable	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ project_info	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ task	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ user_info	★ Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
4 tables	Sum	0	InnoDB	utf8mb4_general_ci	64.0 KiB	0 B

Figure 2.1 Database Structure

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	user_id	int(11)			No	None		AUTO_INCREMENT	✎ Change ➡ Drop More
☐ 2	username	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 3	email	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 4	password	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 5	user_type	int(11)			Yes	NULL			✎ Change ➡ Drop More
☐ 6	first_name	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 7	last_name	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 8	cp_number	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 9	date_of_birth	date			Yes	NULL			✎ Change ➡ Drop More
☐ 10	address	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More

Figure 2.2 User Info Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	project_id	int(11)			No	None		AUTO_INCREMENT	✎ Change ➡ Drop More
☐ 2	title	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 3	details	text	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 4	image_dir	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 5	status	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 6	created_at	timestamp			No	current_timestamp()			✎ Change ➡ Drop More
☐ 7	due_date	date			Yes	NULL			✎ Change ➡ Drop More
☐ 8	client_name	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 9	client_email	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More
☐ 10	qr_code	varchar(255)	utf8mb4_general_ci		Yes	NULL			✎ Change ➡ Drop More

Figure 2.3 Project Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 deliverable_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 project_id	int(11)			Yes	NULL			Change Drop More
☐	3 title	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 details	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 status	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 created_at	timestamp			No	current_timestamp()			Change Drop More
☐	7 due_date	date			Yes	NULL			Change Drop More
☐	8 completion_date	date			Yes	NULL			Change Drop More
☐	9 review_status	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	10 assigned_to	int(11)			Yes	NULL			Change Drop More

Figure 2.4 Deliverable Table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 task_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 deliverable_id	int(11)			Yes	NULL			Change Drop More
☐	3 project_id	int(11)			Yes	NULL			Change Drop More
☐	4 title	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	5 details	text	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	6 status	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	7 created_at	timestamp			No	current_timestamp()			Change Drop More
☐	8 due_date	date			Yes	NULL			Change Drop More
☐	9 completion_date	date			Yes	NULL			Change Drop More
☐	10 review_status	varchar(255)	utf8mb4_general_ci		Yes	NULL			Change Drop More

Figure 2.5 Task Table

Queries

(User Info)

- Show the users ordered by their user type

```
8 SELECT username,  
7     CASE  
6         WHEN user_type = 1 THEN 'Architect'  
5         WHEN user_type = 2 THEN 'Senior Architect'  
4         ELSE ''  
3     END AS 'Position'  
2 FROM  
1     user_info;
```

1	+	-----	+	-----	+
1		username		Position	
2	+	-----	+	-----	+
3		hanako_tanaka		Architect	
4		ryota_yamamoto		Senior Architect	
5		naomi_suzuki		Architect	
6	+	-----	+	-----	+

- Show the count of each user type

```
10 SELECT  
9     CASE  
8         WHEN user_type = 1 THEN 'Architect'  
7         WHEN user_type = 2 THEN 'Senior Architect'  
6         ELSE ''  
5     END AS 'Position',  
4     COUNT(*) AS Count  
3 FROM user_info  
2 GROUP BY user_type;
```

- Show the users without a cellphone number

```
1 SELECT username, first_name, last_name FROM user_info WHERE cp_number IS NULL;
```

1	+	-----	+	-----	+	-----	+
1		username		first_name		last_name	
2	+	-----	+	-----	+	-----	+
3		hanako_tanaka		Hanako		Tanaka	
4	+	-----	+	-----	+	-----	+

- Selecting the contact information of all the users

```
1 SELECT username, cp_number, email from 'user_info'
```

1	+	-----	+	-----	+
1		username		cp_number	
2		email			
2	+	-----	+	-----	+
3		hanako_tanaka		NULL	
4		ryota_yamamoto		09876543210	
5		naomi_suzuki		09187654321	
6	+	-----	+	-----	+

- Login credentials, selecting the user id from a given username and password

```
1 SELECT user_id as 'Login Credentials' FROM user_info WHERE password =  
'naomipass';
```

1	+	-----	+
1		Login Credentials	
2	+	-----	+
3		3	
4	+	-----	+

(Project)

- Show the projects in order of the closes due date

```
7 SELECT  
6 | title as 'Project',  
5 | due_date  
4 FROM  
3 | project_info  
2 ORDER BY  
1 | due_date
```

1	+	-----	+	-----	+
1		Project		due_date	
2	+	-----	+	-----	+
3		Path of Colors		2024-06-30	
4		Tree of Life Project		2024-07-15	
5		Dream Treasure		2024-09-30	
6	+	-----	+	-----	+

- Show the directory of the reference image of each project with its accompanying details

```
6 SELECT
5 | title as Project,
4 | details,
3 | image_dir
2 FROM
1 | project_info
```

1	title	details	image_dir
2			
3	Tree of Life Project	Detailed plan for the project	/images/tree_of_life.jpg
4	Dream Treasure	Details for building the dream house	/images/dream_treasure.jpg
5	Path of Colors	Blueprint for the innovative project	/images/path_of_colors.jpg
6			

- Show the QR code of each project with its accompanying details

```
3 SELECT
2 | title as Project,
1 | details,
4 | qr_code
1 FROM
2 | project_info
```

1	Project	details	qr_code
2			
3	Tree of Life Project	Detailed plan for the project	ABC123
4	Dream Treasure	Details for building the dream house	DEF456
5	Path of Colors	Blueprint for the innovative project	GHI789
6			

- Show the project information based on a given client email

```
1 SELECT
1 | title as Project,
2 | details, client_name
3 FROM
4 | project_info
5 WHERE
6 | client_email = 'mario.yamazaki@example.com'
```

1	+	-----	+	-----	+	-----	+
1		Project		details		client_name	
2	+	-----	+	-----	+	-----	+
3		Tree of Life Project		Detailed plan for the project		Mario Yamazaki	
4	+	-----	+	-----	+	-----	+

- Show the projects that are due in a specified month

```

9 SELECT
8 |   title as Project,
7 |   details,
6 |   due_date
5 FROM
4 |   project_info
3 WHERE
2 |   YEAR(due_date) = 2024 AND
1 |   MONTH(due_date) = 7
10 |

```

1	+	-----	+	-----	+	-----	+
1		Project		details		due_date	
2	+	-----	+	-----	+	-----	+
3		Tree of Life Project		Detailed plan for the project		2024-07-15	
4	+	-----	+	-----	+	-----	+

(Deliverable)

- Show the deliverables under the project ordered by the closest due date.

```

1 |SELECT (SELECT title FROM project_info WHERE due_date = (SELECT MIN(due_date)
FROM project_info)) as 'Project',
1 |title as 'Deliverable', due_date as 'Deliverable Deadline'
2 |FROM deliverable
3 |WHERE project_id = (SELECT project_id FROM project_info WHERE due_date =
(SELECT MIN(due_date) FROM project_info))
4 |ORDER BY due_date;

```

1	+	-----	+	-----	+	-----	+
1		Project		Deliverable		Deliverable Deadline	
2	+	-----	+	-----	+	-----	+
3		Path of Colors		Project Prototype		2024-06-15	
4		Path of Colors		Final Report		2024-06-30	
5	+	-----	+	-----	+	-----	+

- Show the completed deliverables and its description

```

1 SELECT title as 'Completed Deliverable', details as 'Description' FROM
deliverable WHERE status = 'Completed';

```


1	Completed Deliverable	Description
2		
3	Project Prototype	First version of project prototype
4	Final Report	Comprehensive report on project outcomes
5		

- Show the deliverables to be completed on the month of July and its accompanying description and deadline

```

10 SELECT
9   title as 'Deliverable To Be Completed in July',
8   details,
7   due_date
6 FROM
5   deliverable
4 WHERE
3   YEAR(due_date) = 2024
2 AND
1   MONTH(due_date) = 7

```

1	Deliverable To Be Completed in July	details	due_date
2			
3	Initial Design Proposal	Draft proposal for the project	2024-07-01
4	Client Feedback Report	Document summarizing client feedback	2024-07-15
5			

- Show every deliverable (with Project Title) that has not been completed with its title and due date

```

7 SELECT
6   (SELECT p.title FROM project_info p WHERE p.project_id = d.project_id)
5   as Project,
4   d.title, due_date, status
3 FROM deliverable d
2 WHERE
1   d.status != 'Completed';

```

1	Project	title	due_date	status
2				
3	Tree of Life Project	Initial Design Proposal	2024-07-01	Pending
4	Tree of Life Project	Client Feedback Report	2024-07-15	In progress
5	Dream Treasure	Architectural Blueprint	2024-08-15	In progress
6	Dream Treasure	Material Selection List	2024-09-15	Pending
7				

- Show every deliverable that has been rejected

```

1 +-----+
1 | Deliverable | details |
2 +-----+
3 | Final Report | Comprehensive report on project outcomes |
4 +-----+

```

- Show every task that has been accepted in its review phase

```

3 SELECT title as 'Accepted Tasks', details, completion_date
2 FROM task
1 WHERE review_status = "Accepted"

```

	Accepted Tasks	details	completion_date
1	Prototype Development	Develop the project prototype	2024-06-30
2	Prototype Testing	Test the functionality of the prototype	2024-07-10

- Show every task where the due date is after the month of June

```

2 SELECT title as 'Accepted Tasks', details, due_date
1 FROM task
3 WHERE MONTH(due_date) > 6 AND YEAR(due_date) = 2024
1

```

	Tasks, Deadline After June	details	due_date
1	Compile Client Feedback	Compile feedback from the client	2024-07-10
2	Review Feedback	Review and incorporate client feedback	2024-07-15
3	Architectural Planning	Plan the architectural layout	2024-07-20
4	Material Sourcing	Source materials for the project	2024-08-01
5	Material Procurement	Procure materials for construction	2024-08-15
6	Finalize Selection	Finalize selection of materials	2024-08-30
7	Prototype Testing	Test the functionality of the prototype	2024-07-01
8	Report Compilation	Compile a comprehensive project report	2024-07-15
9	Report Review	Review the project report for accuracy	2024-07-21

- Count the number of tasks that have been accepted and rejected in its review phase

```

2 SELECT COUNT(review_status) as 'Count by Review Status', review_status
1 FROM task WHERE review_status IS NOT NULL GROUP BY review_status;

```

	Count by Review Status	review_status
1	2	Accepted
2	2	Pending

Data inside each table for reference

user_info table

		user_id	username	email	password	user_type	first_name	last_name	cp_number	date_of_birth	address
☐	 Edit  Copy  Delete	1	hanako_tanaka	hanako.tanaka@example.com	password123	1	Hanako	Tanaka	NULL	1990-05-15	123 Mabini Street
☐	 Edit  Copy  Delete	2	ryota_yamamoto	ryota.yamamoto@example.com	securepass	2	Ryota	Yamamoto	09876543210	1988-10-20	456 Quezon Avenue
☐	 Edit  Copy  Delete	3	naomi_suzuki	naomi.suzuki@example.com	naomipass	1	Naomi	Suzuki	09187654321	1995-03-25	789 Rizal Boulevard

project_info table

		project_id	title	details	image_dir	status	created_at	due_date	client_name	client_email	qr_code
☐	 Edit  Copy  Delete	1	Tree of Life Project	Detailed plan for the project	/images/tree_of_life.jpg	In progress	2024-05-05 22:39:03	2024-07-15	Mario Yamazaki	mario.yamazaki@example.com	ABC123
☐	 Edit  Copy  Delete	2	Dream Treasure	Details for building the dream house	/images/dream_treasure.jpg	Pending	2024-05-05 22:39:03	2024-09-30	Naza Nagasaki	naza.nagasaki@example.com	DEF456
☐	 Edit  Copy  Delete	3	Path of Colors	Blueprint for the innovative project	/images/path_of_colors.jpg	Completed	2024-05-05 22:39:03	2024-06-30	Wasa Kawasaki	wasa.kawasaki@example.com	GHI789

deliverable table

		deliverable_id	project_id	title	details	status	created_at	due_date	completion_date	review_status	assigned_to
☐	 Edit  Copy  Delete	1	1	Initial Design Proposal	Draft proposal for the project	Pending	2024-05-06 00:53:48	2024-07-01	NULL	NULL	1
☐	 Edit  Copy  Delete	2	1	Client Feedback Report	Document summarizing client feedback	In progress	2024-05-06 00:53:48	2024-07-15	NULL	NULL	2
☐	 Edit  Copy  Delete	3	2	Architectural Blueprint	Detailed architectural plan	In progress	2024-05-06 00:53:48	2024-08-15	NULL	NULL	1
☐	 Edit  Copy  Delete	4	2	Material Selection List	List of materials for construction	Pending	2024-05-06 00:53:48	2024-09-15	NULL	NULL	2
☐	 Edit  Copy  Delete	5	3	Project Prototype	First version of project prototype	Completed	2024-05-06 00:53:48	2024-06-15	2024-06-30	Accepted	1
☐	 Edit  Copy  Delete	6	3	Final Report	Comprehensive report on project outcomes	Completed	2024-05-06 00:53:48	2024-06-30	2024-07-15	Rejected	2

task table

		task_id	deliverable_id	project_id	title	details	status	created_at	due_date	completion_date	review_status
☐	 Edit  Copy  Delete	1	1	1	Initial Research	Conduct initial research for the proposal	Pending	2024-05-06 00:55:10	2024-06-15	NULL	NULL
☐	 Edit  Copy  Delete	2	1	1	Draft Proposal	Create the initial draft proposal document	In progress	2024-05-06 00:55:10	2024-06-30	NULL	NULL
☐	 Edit  Copy  Delete	3	2	1	Compile Client Feedback	Compile feedback from the client	Pending	2024-05-06 00:55:10	2024-07-10	NULL	NULL
☐	 Edit  Copy  Delete	4	2	1	Review Feedback	Review and incorporate client feedback	In progress	2024-05-06 00:55:10	2024-07-15	NULL	NULL
☐	 Edit  Copy  Delete	5	3	2	Architectural Planning	Plan the architectural layout	In progress	2024-05-06 00:55:10	2024-07-20	NULL	NULL
☐	 Edit  Copy  Delete	6	3	2	Material Sourcing	Source materials for the project	Pending	2024-05-06 00:55:10	2024-08-01	NULL	NULL
☐	 Edit  Copy  Delete	7	4	2	Material Procurement	Procure materials for construction	In progress	2024-05-06 00:55:10	2024-08-15	NULL	NULL
☐	 Edit  Copy  Delete	8	4	2	Finalize Selection	Finalize selection of materials	Pending	2024-05-06 00:55:10	2024-08-30	NULL	NULL
☐	 Edit  Copy  Delete	9	5	3	Prototype Development	Develop the project prototype	Completed	2024-05-06 00:55:10	2024-06-20	2024-06-30	Accepted
☐	 Edit  Copy  Delete	10	5	3	Prototype Testing	Test the functionality of the prototype	Completed	2024-05-06 00:55:10	2024-07-01	2024-07-10	Accepted
☐	 Edit  Copy  Delete	11	6	3	Report Compilation	Compile a comprehensive project report	Completed	2024-05-06 00:55:10	2024-07-15	2024-07-20	Pending
☐	 Edit  Copy  Delete	12	6	3	Report Review	Review the project report for accuracy	Completed	2024-05-06 00:55:10	2024-07-21	2024-07-25	Pending